

Cambridge (CIE) IGCSE Physics
Paper 1: Multiple Choice (Set E)
Core

Saturday 26 April 2025

Afternoon (Time: 0 hours 45 minutes)

Total marks

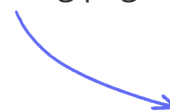
/ 40

Instructions

- Try to complete this mock exam paper in one sitting, under exam conditions. Use all the time available and check your answers to each question at the end before submitting.
- Remember this is PRACTICE. Mistakes are fine and will help you improve in time for the real exam - just do your best.
- You may use a calculator.
- Take the weight of 1.0 kg to be 9.8N (acceleration of free fall = 9.8m/s^2).

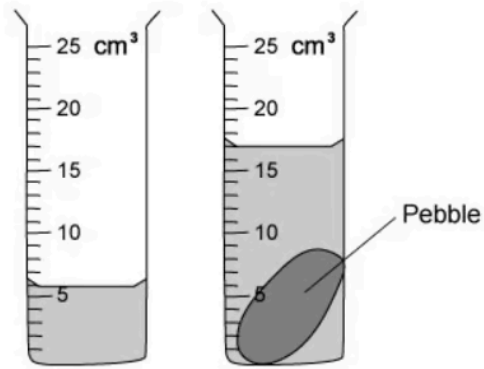
Scan here to mark your mock exam

or visit the mock exams landing page for this course



- 1 A geologist wants to measure the volume of a particularly interesting pebble she has found in a river.

She uses the apparatus shown below.

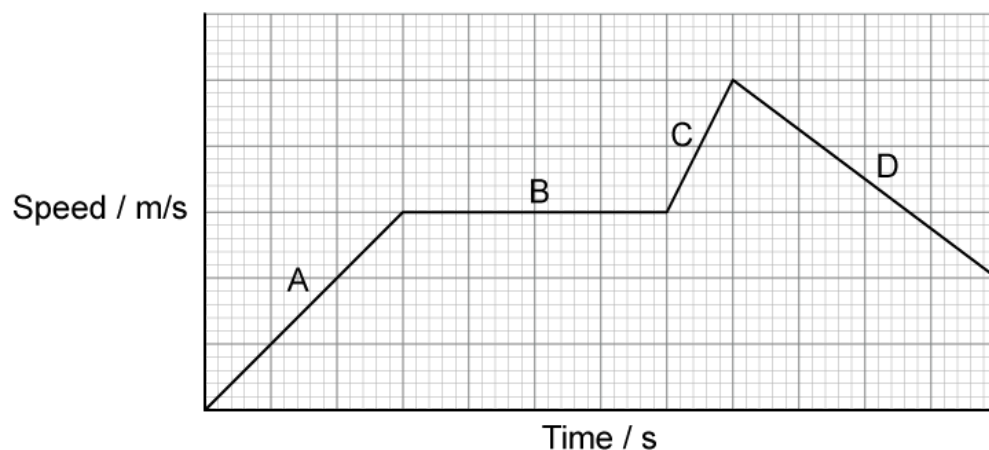


What is the volume of the pebble?

- A. 11 cm³
- B. 23 cm³
- C. 6 cm³
- D. 17 cm³

(1 mark)

- 2 In which section of the graph is the most distance covered?



(1 mark)

- 3** During a Go Karting race, a car does 8 laps of a 300 m course. It takes 4.0 minutes to complete the race.

What was the average speed of the Go Kart?

- A.** 75 m/s
- B.** 10 m/s
- C.** 1.25 m/s
- D.** 600 m/s

(1 mark)

- 4** A cheesemonger uses a newton meter to measure the weight of a block of cheese.

The cheese has a weight of 5.0 N

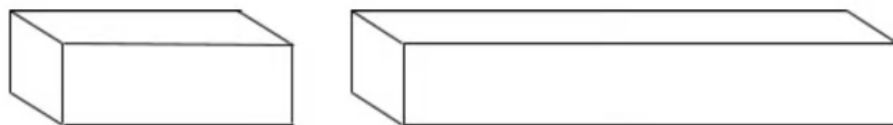
What is the approximate mass of the cheese?

- A.** 0.05 kg
- B.** 0.5 kg
- C.** 5 kg
- D.** 50 kg

(1 mark)

- 5** Two beams have the same rectangular cross section, but one is longer than the other.

Both beams are made from the same material.



Which quantity is the same in both beams?

- A.** The volume
- B.** The weight

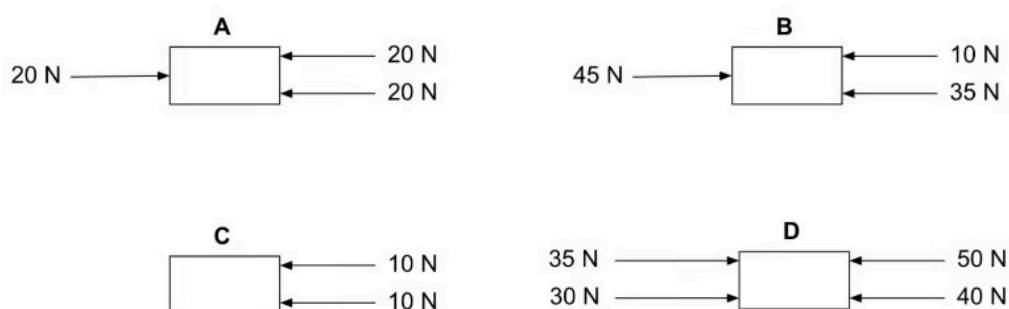
C. The mass

D. The density

(1 mark)

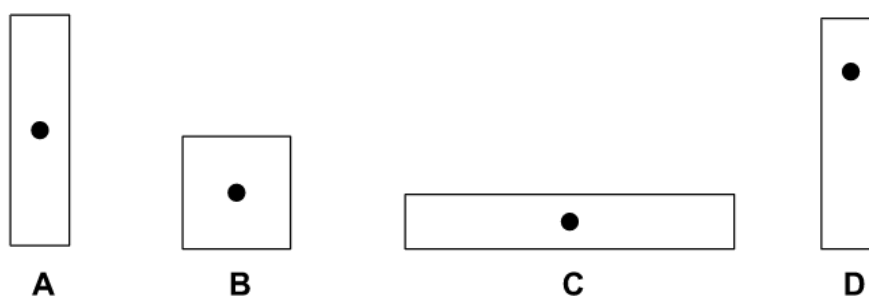
- 6 The diagram shows four objects, each with the same mass. Each object is acted on by the forces shown.

Which object will experience the largest acceleration to the left?



(1 mark)

- 7 Four cuboids are shown in the diagram below. The position of their centre of gravity is also shown.

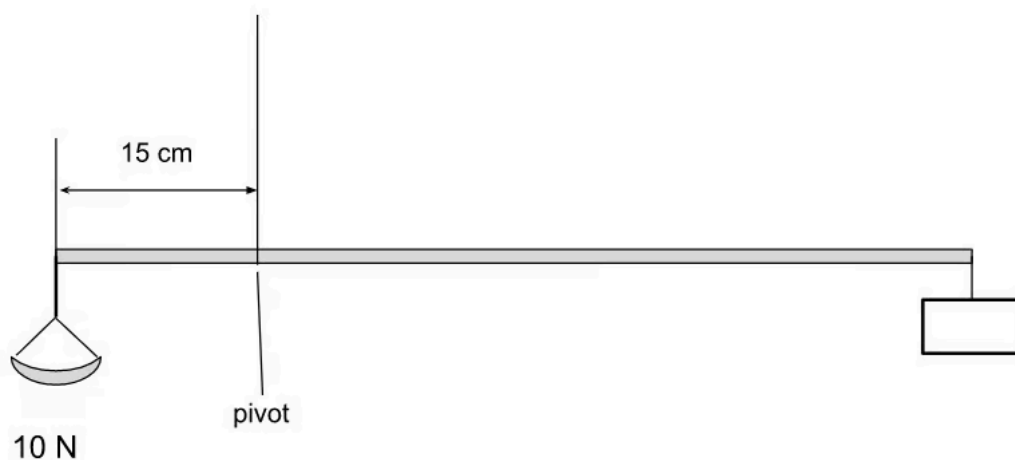


Which of the cuboids is the most stable?

(1 mark)

- 8 A 'yardarm' is a stick, hung from a string, that is used to weigh goods. Goods are placed in the pan on the left, and weights are added to the box on the right side until it

balances. The weight of the pan and the box are such that the weight of the yardarm itself is entirely balanced by the pan. The yardarm is 65 cm long.



How many 0.5 N weights need to be added to the box in order to balance the yardarm?

- A. 2
- B. 3
- C. 6
- D. 12

(1 mark)

- 9 A trolley is pushed with a force that causes it to move a certain distance. It moves in the direction it is pushed.

Which row in the table represents the combination of force and distance that means the smallest amount of work is done on the trolley?

	Force / N	Distance moved / m
A	20	20
B	40	10
C	50	6.0
D	70	4.0

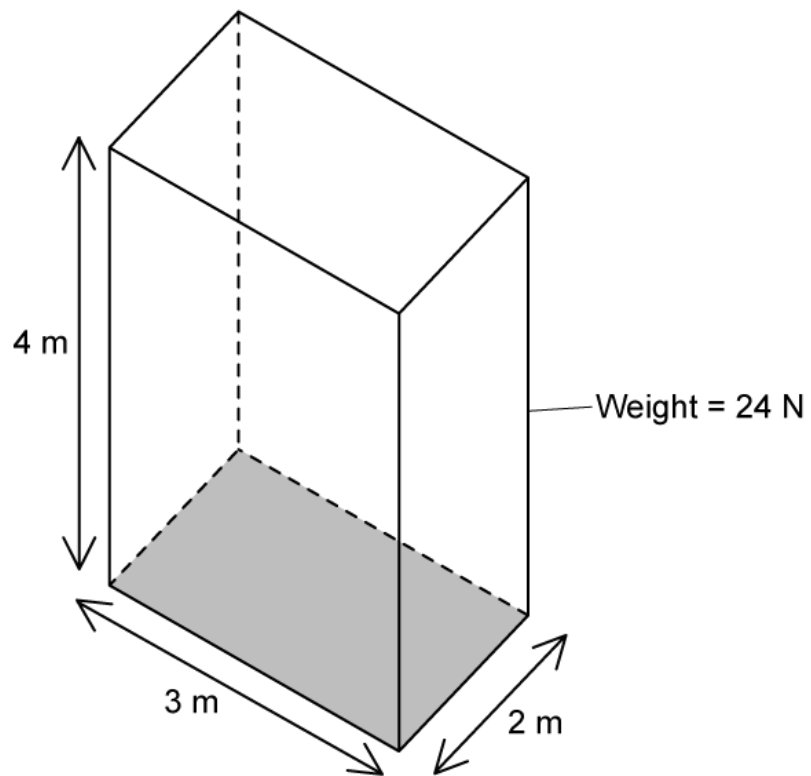
(1 mark)

10 Which energy resource does not release greenhouse gases?

- A.** Oil
- B.** Hydro-electric power
- C.** Biofuel
- D.** Coal

(1 mark)

11 What pressure does the object in the diagram below exert on the ground beneath it?

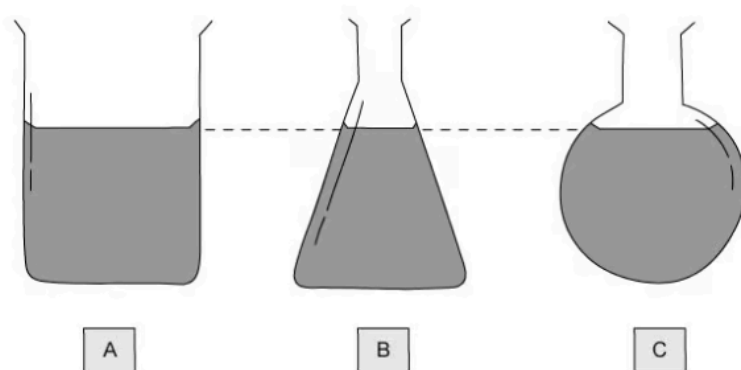


- A.** 4.0 Pa
- B.** 6.0 Pa
- C.** 12 Pa
- D.** 24 Pa

(1 mark)

12 Three beakers of water are placed on a table. The depth of water in each container is the

same.



In which container does the water exert the greatest pressure on the base of the container?

- A.** A
- B.** B
- C.** C
- D.** None, the pressure is the same in all three.

(1 mark)

13 Descriptions of two different states of matter are shown below.

To which state of matter do each of the descriptions fit?

X: the molecules are very close together, in fixed positions. They vibrate around these fixed positions.

Y: the molecules move rapidly and randomly in straight lines until they collide with something. The molecules are spaced apart very far.

	State X	State Y
A	solid	liquid
B	solid	gas
C	liquid	gas
D	gas	liquid

(1 mark)

- 14** A sealed yet empty (except for air) plastic drinks bottle is left outside in the sunshine on a warm day.

Which row of the table correctly describes what happens to the average speed of the air molecules, and to the pressure inside the bottle?

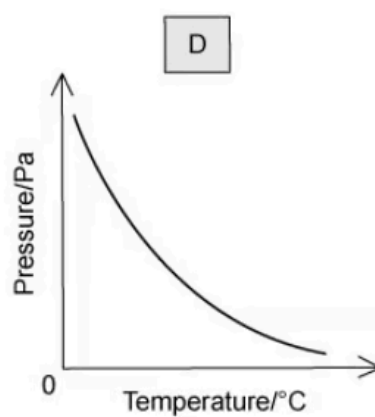
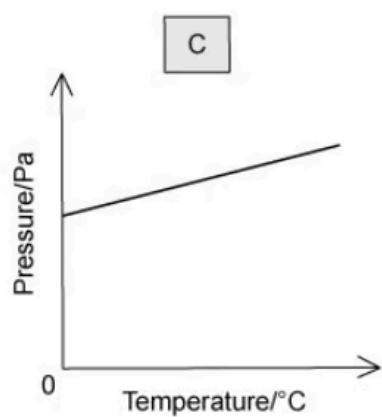
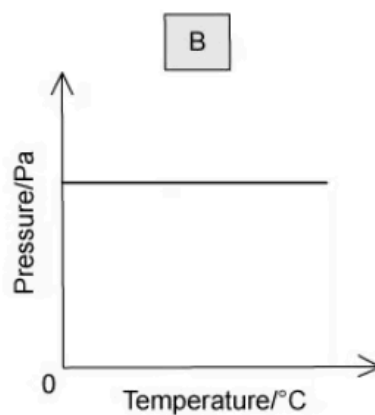
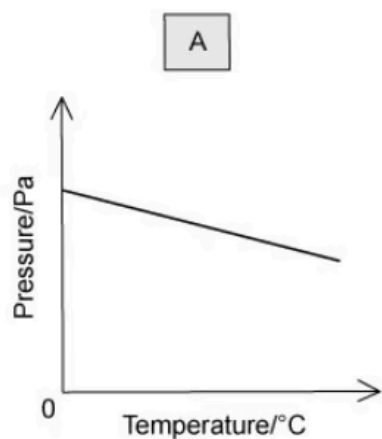
	Average speed of particles	Pressure inside bottle
A	increases	increases
B	increases	decreases
C	decreases	increases
D	decreases	decreases

(1 mark)

15 Air is trapped inside a cylinder. The volume of the cylinder cannot change.

The temperature of the air is increased.

Which of the graphs below shows the correct relationship between pressure and temperature?



(1 mark)

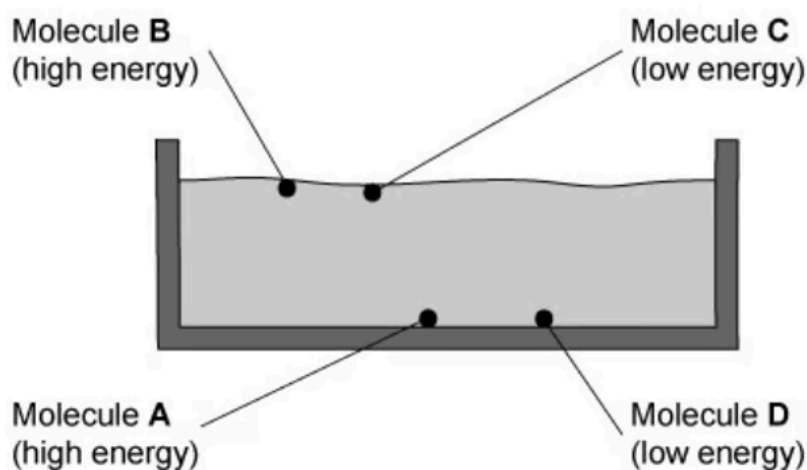
16 Which statement best describes a metal experiencing thermal expansion?

- A.** The size of the metal atoms increases
- B.** The space between the metal atoms increases
- C.** The metal atoms change state
- D.** The metal atoms gain enough energy to overcome the intermolecular forces of attraction holding them in place

(1 mark)

17 The diagram shows four water molecules in a swimming pool. It also shows how much energy each molecule has.

Which of the water molecules is most likely to *evaporate* from the liquid?

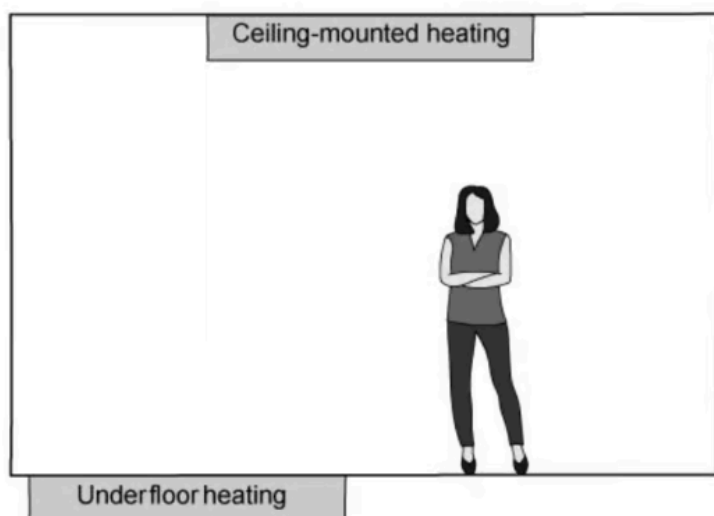


- A.** Molecule A
- B.** Molecule B
- C.** Molecule C
- D.** Molecule D

(1 mark)

18 In modern housing, it is convenient to locate heaters so they are not on the walls, taking up valuable wall space for furniture.

Other than wall-mounted radiators, two other options would be viable: ceiling heaters or underfloor heaters.



Which of the two options would heat the room more effectively and why?

	Option	Explanation
A	ceiling-mounted	hot air is more dense and falls
B	ceiling-mounted	hot air is less dense and falls
C	underfloor	hot air is less dense and rises
D	underfloor	hot air is more dense and rises

(1 mark)

- 19** A student suggests some uses for the knowledge that black and white objects emit and absorb different amounts of infrared radiation.

Which of her suggestions is correct?

- A.** Coffee mugs should be black to keep the coffee hot.
- B.** Houses in hot countries should be painted white to keep the occupants cool.
- C.** White clothing will help you cool down much faster.
- D.** Solar heating panels should be white to heat the water more quickly.

(1 mark)

- 20** What is the unit of wave speed?

- A.** Hertz
- B.** Metre
- C.** Second
- D.** Metres per second

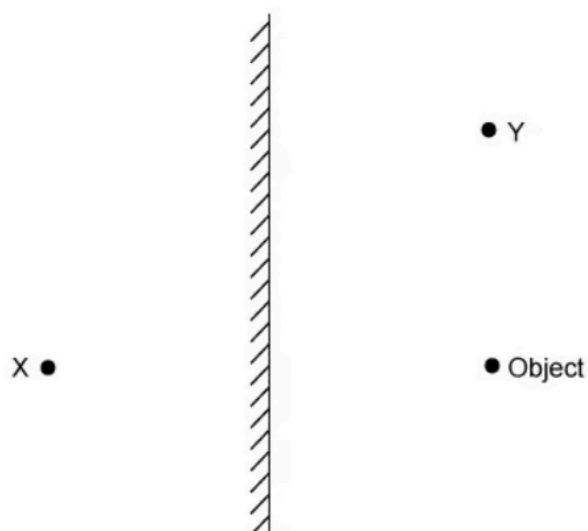
(1 mark)

- 21** Water waves travel from deeper water to more shallow water. This causes them to refract.

Which wave property always remains the same when refraction occurs?

- A.** Wavelength
- B.** Frequency
- C.** Amplitude
- D.** Speed

(1 mark)



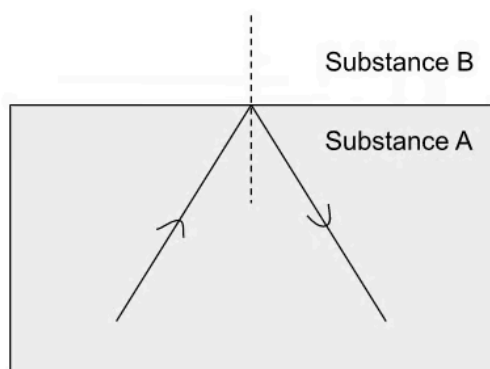
An object is placed in front of a mirror. Where is the image formed, and is the image real or virtual?

	Image location	Image type
A	X	real
B	X	virtual
C	Y	real
D	Y	virtual

(1 mark)

23 The diagram shows a ray of light being totally internally reflected at the boundary

between substances A and B.



Which row in the table must be true?

	Angle of incidence	Substance A
A	greater than critical angle	greater refractive index than B
B	greater than critical angle	smaller refractive index than B
C	less than critical angle	greater refractive index than B
D	less than critical angle	smaller refractive index than B

(1 mark)

24 In what way are microwaves different from ultraviolet waves?

- A.** Ultraviolet waves travel faster than microwaves in a vacuum.
- B.** Microwaves have a longer wavelength than ultraviolet waves.
- C.** Microwaves have a higher frequency than ultraviolet waves.
- D.** Microwaves are longitudinal while ultraviolet waves are transverse.

(1 mark)

25 The table gives common uses of electromagnetic radiation.

Which row correctly matches the type of electromagnetic radiation with its use?

	Detecting banknote forgery	TV remote controllers	Sterilising medical instruments
A	microwaves	visible light	microwaves
B	infrared	infrared	γ-rays
C	ultraviolet	radio waves	infrared
D	ultraviolet	infrared	γ-rays

(1 mark)

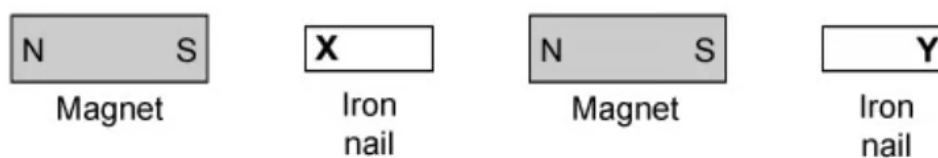
26 Which of the following choices of metal would be suitable for:

- Use as a permanent magnet?
- Use as the core in an electromagnet?

	Use as a permanent magnet	Use as the core of an electromagnet
A	steel	iron
B	steel	steel
C	iron	iron
D	iron	steel

(1 mark)

- 27** A student lines up some permanent magnets with soft iron nails in between as shown in the diagram below.



What would the magnetic poles of locations **X** and **Y** be?

	X	Y
A	north	north
B	north	south
C	south	north
D	south	south

(1 mark)

28 Which pair of quantities have the same units?

- A. Current and potential difference
- B. Current and resistance
- C. Electromotive force and potential difference
- D. Electromotive force and resistance

(1 mark)

29 A student wants to charge up some rods by rubbing them with a cloth made from a suitable material.

He has a number of rods to choose from: a copper rod, a PVC rod, an aluminium rod and a glass rod.

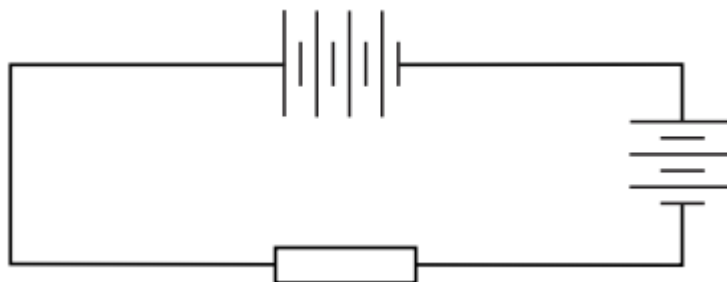
Which two of the rods would hold a charge when rubbed with an appropriate cloth?

- A. Aluminium and PVC
- B. Glass and copper
- C. Aluminium and copper
- D. Glass and PVC

(1 mark)

30 The circuit diagram shows cells connected with a resistor in a series circuit.

Each cell has an electromotive force (e.m.f.) of 1.5 V



What is the total e.m.f. in the circuit?

- A.** 1.5 V
- B.** 4.5 V
- C.** 6.0 V
- D.** 10.5 V

(1 mark)

31 The plug for an electric lamp contains a 3 A fuse. The fuse's job is to protect the lamp if too much current flows.

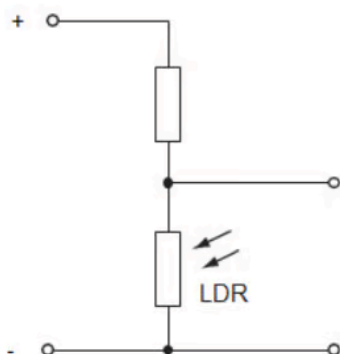
Where is the fuse positioned, and what happens when the fuse does its job?

	Where is it positioned?	What does it do?
A	live wire	reduces the current to 3 A
B	neutral wire	breaks the circuit
C	live wire	breaks the circuit
D	neutral wire	reduces the current to 3 A

(1 mark)

32 A homeowner buys a security light that switches on automatically when it gets dark.

The security light contains the circuit, shown below.



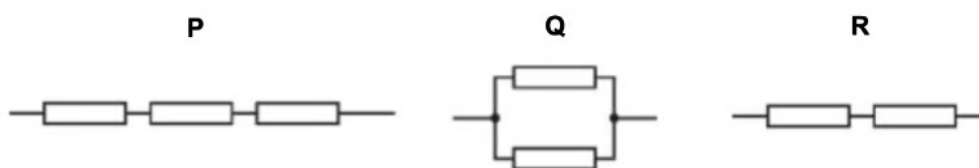
When it gets dark, what happens to the resistance of the LDR, and what happens to the potential difference across the LDR?

	Resistance of LDR	p.d. across LDR
A	increases	increases
B	decreases	increases
C	increases	decreases
D	decreases	decreases

(1 mark)

33 A student connects together some identical resistors in three different combinations, as shown in the diagram.

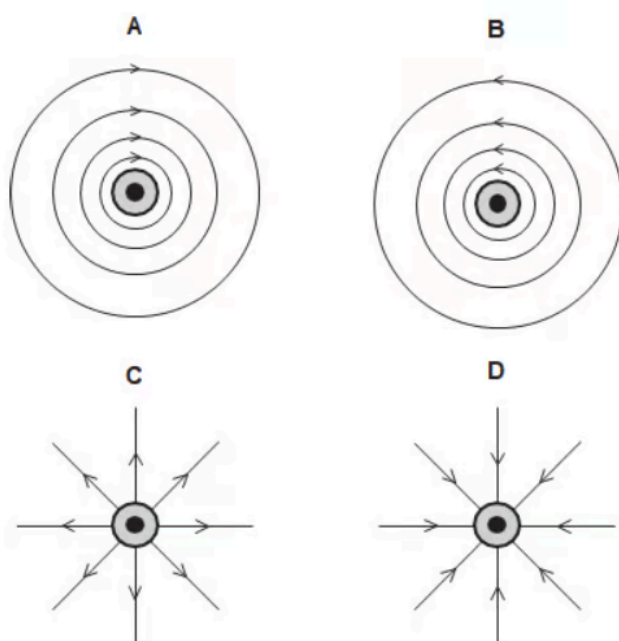
Which of the answers below shows the correct order of the combinations from the lowest resistance to the highest?



- A.** $P \rightarrow Q \rightarrow R$
- B.** $Q \rightarrow P \rightarrow R$
- C.** $Q \rightarrow R \rightarrow P$
- D.** $P \rightarrow R \rightarrow Q$

(1 mark)

- 34** Which diagram correctly shows the magnetic field lines around a current-carrying wire if the current is directed out of the page?



(1 mark)

- 35** The number of turns in the primary coil and secondary coil of a transformer is N_p and N_s respectively.

Which of the following statements represents a step-up transformer?

A. $N_s = N_p$

B. $N_s > N_p$

C. $N_s < N_p$

D. $N_p > N_s$

(1 mark)

- 36** The nuclide notation for a generic element is given below:



Which of the following is the correct definition of each of the terms?

	X	A	Z
A	symbol	nucleon number	proton number
B	symbol	proton number	nucleon number
C	symbol	neutron number	proton number
D	nucleon number	proton number	neutron number

(1 mark)

37 Which statement is **not** a method used to minimise the risk caused by working with radioactive sources.

- A.** Store the sources in lead-lined boxes.
- B.** Minimise the amount of time spent handling the sources.
- C.** Keep the source cold.
- D.** Keep the source as far away as possible, for example, using a pair of tongs.

(1 mark)

38 The count rate from a radioactive isotope is recorded every hour. The count rate is corrected for background radiation. The table shows the readings.

time / hours	0	1	2	3	4	5
corrected count rate / counts/s	1200	990	816	673	555	458

What estimate of the half-life of the isotope can be obtained from the readings in the table?

- A.** between 1 and 2 hours
- B.** between 2 and 3 hours
- C.** between 3 and 4 hours
- D.** between 4 and 5 hours

(1 mark)

- 39** The radius of Jupiter's orbit is 778×10^6 km. What is the time taken for light from the Sun to reach the surface of Jupiter?

The speed of light is 3×10^8 m/s.

- A.** 2600 ms
- B.** 2600 s
- C.** 0.39 s
- D.** 3.9×10^{-4} s

(1 mark)

- 40** Which two elements are the most commonly found in the Sun?

- A.** Hydrogen and oxygen
- B.** Helium and carbon
- C.** Hydrogen and helium
- D.** Nitrogen and oxygen

(1 mark)